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#include <stdio.h>
#include <ctype.h>
#include <string.h>

int main()
{
    char s[500];
    int i = 0;

    printf("Enter a line of code:\n");
    fgets(s, sizeof(s), stdin);

    printf("\nTokens:\n");

    while (s[i] != '\0')
    {
        /* Ignore spaces, tabs and new lines */
        if (isspace(s[i]))
        {
            i++;
            continue;
        }

        /* Ignore single-line comments */
        if (s[i] == '/' && s[i + 1] == '/')
            break;

        /* Ignore multi-line comment */
        if (s[i] == '/' && s[i + 1] == '*')
        {
            i += 2;

            while (s[i] != '\0' &&
                !(s[i] == '*' && s[i + 1] == '/'))
                i++;

            if (s[i] != '\0')
                i += 2;

            continue;
        }

        /* Identifier */
        if (isalpha(s[i]) || s[i] == '_')
        {
            printf("IDENTIFIER: ");

            while (isalnum(s[i]) || s[i] == '_')
            {
                printf("%c", s[i]);
                i++;
            }

            printf("\n");
            continue;
        }

        /* Constant */

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    if (isdigit(s[i]))
    {
        printf("CONSTANT: ");

        while (isdigit(s[i]))
        {
            printf("%c", s[i]);
            i++;
        }

        printf("\n");
        continue;
    }

    /* Operator */
    if (strchr("+-*/%=<>!", s[i]))
    {
        printf("OPERATOR: %c\n", s[i]);
        i++;
        continue;
    }

    i++;
}

return 0;
}

```

OUTPUT

```

Enter a line of code:
int a = 10;

Tokens:
IDENTIFIER: int
IDENTIFIER: a
OPERATOR: =
CONSTANT: 10

-----
Process exited after 20.77 seconds with return value 0
Press any key to continue . . .

```